

A Residual Attention U-Net Approach for High-Resolution Satellite Image Segmentation

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Abstract—Semantic segmentation of high-resolution satellite images is a critical task in remote sensing, enabling detailed analysis for applications such as urban development, environmental monitoring, and disaster management. This paper introduces a Residual Attention U-Net architecture that integrates residual blocks for effective feature extraction, improving segmentation accuracy. To facilitate focused attention on relevant spatial regions, attention gates are also included. These improvements allow the model to address the challenges of complex and diverse aerial imagery by maintaining essential spatial details and reducing background noise. Our method is evaluated on a high-resolution aerial dataset that includes multiple land cover categories, such as water bodies, roads, buildings, vegetation, and open land areas, allowing for a comprehensive assessment across different types of terrain. The Residual Attention U-Net achieved strong performance across multiple evaluation metrics including Accuracy and Intersection over Union (IoU), with an overall IoU score of 81.07% and an accuracy of 92.33%. Comparative analysis with other U-Net variants demonstrates the superior capability of our model to capture intricate details and boundaries within satellite images, making it a valuable tool for precise segmentation in remote sensing applications. These findings underscore the potential of the Residual Attention U-Net in providing high-accuracy segmentation for high-resolution satellite imagery, paving the way for its use in various geospatial and environmental studies.

Index Terms—Satellite Image, Computer vision, Semantic Segmentation, Deep Learning, Residual U-Net, Intersection over Union(IoU).

I. INTRODUCTION

Aerial image segmentation has become an essential task in various applications, including urban planning, environmental monitoring, and disaster management. The ability to accurately segment objects from satellite images is crucial for extracting meaningful insights and supporting decision-making processes. Convolutional neural networks (CNNs) [1], and specifically the U-Net architecture [2], have gained popularity in this domain due to their encoder-decoder structure, which enables precise localization of features and effective handling of spatial information. However, traditional U-Net models often face limitations in complex scenes where detailed feature extraction is necessary, particularly in high-resolution aerial images with diverse and intricate landscapes. Pixel-wise semantic segmentation accurately discerns and interprets object appearance and dividing lines, demonstrating spatial correlation between elements within a single image, which can be visualized in Fig. 1.

Residual U-Net models, an extension of the original U-Net, introduce residual blocks to improve gradient flow and facilitate better feature propagation, addressing issues such as vanishing gradients and limited feature reuse in deeper networks. This residual learning approach [3] has proven effective in enhancing segmentation accuracy, yet there is still room for improvement, especially in terms of focusing on critical regions of interest within aerial images.

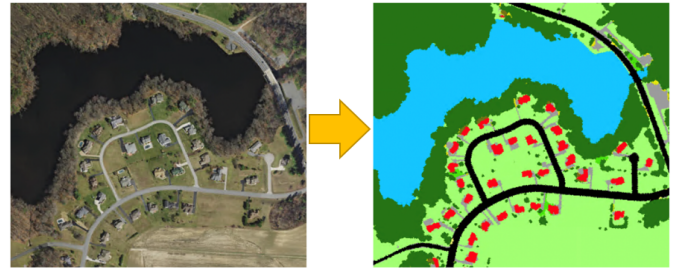


Fig. 1: An example of satellite image and its semantic segmentation for different classes (Blue for water body, Red for buildings, etc).

To further advance segmentation performance, attention mechanisms [4] have been introduced to allow models to concentrate on salient regions in the feature space, reducing the influence of irrelevant or noisy information. Integrating attention mechanisms with residual U-Net architectures, particularly through attention gates, can potentially improve model performance by emphasizing relevant features and refining the segmentation output.

The primary contributions of the proposed methodology can be stated as follows:

- 1) We developed an effective pre-processing pipeline for high-resolution satellite images to improve input quality.
- 2) We implemented modified residual connections for effective feature extraction, while the attention gates selectively focus on important spatial regions, improving segmentation accuracy.
- 3) We validated our approach by surpassing traditional U-Net and existing U-Net variants, achieving a superior IoU score.

This paper develops our proposal in a methodical manner. Section 4 presents our system, Section 3 examines used

datasets, and Section 2 analyzes current semantic segmentation techniques. While Section 6 provides a brief summary of our findings, Section 5 evaluates its effectiveness.

II. LITERATURE REVIEW

Semantic segmentation in high-resolution imagery relies on models capable of extracting both detailed spatial information and high-level abstractions such as object boundaries, contextual relationships, and class-specific features from data. Research [5] presents a CNN-based approach that employs a downsampling-then-upsampling architecture to perform pixel-level land-cover classification in satellite images. This architecture first captures a rough spatial map of high-level features through downsampling, then refines these representations by upsampling to the original resolution. Similarly, [6] extends DenseNets use a downsampling path to extract coarse semantic features and an upsampling path to restore the input resolution, improving feature reuse across layers.

Deep learning-based semantic segmentation techniques have shown superior performance to conventional machine learning and statistical approaches mentioned in the study [1]. Among these, U-Net stands out by achieving the highest IoU score, largely due to its encoder-decoder structure with skip connections. The authors in [7] presented deep learning techniques for semantic segmentation, demonstrating that U-Net outperformed the PSPNet architecture with a Mean-IoU score of 0.51. Ming et al. [8] implemented attention-dilated LinkNet for segmentation tasks, where LinkNet is a U-Net with ResNet in its downsampling path.

Traditional methods like AlexNet [9] employ neuron layer output to the next neuron layer, which excludes the low-level representation from the last layer, which serves as the decision level, even though new concepts enable a neuron layer to summarize the information from many more prior layers. Because ResNet components fuse the output representation of a block of 2D convolution filters with the input, Residual Networks (ResNet) [3] allow mixing input data with a new slightly higher representation level. In [10], ResNet is also utilized as the encoder of an FPN network. The SegNet architecture [11] employs an auto-encoder structure where encoder captures multi-scale features by progressively reducing the spatial resolution and decoder then upsamples these features.

Semantic segmentation has been widely utilized across various sectors, including applications mentioned in monitoring deforestation [12], ship detection [13], and road detection [14]. In each application, the U-Net architecture has been effectively employed to perform these tasks with high precision. Khan et al. [15], Aburaed et al. [16], Patil et al. [17] and Maithil et al. [18] modified the U-Net architecture to enhance the segmentation of satellite images from the [19] dataset.

Given that segmentation in satellite images is a prerequisite for encoder-decoder-based architectures, U-Net meets this requirement with its phase-simple design and ease of implementation, allowing for straightforward modification. Incorporating residual blocks could further enhance its ability to accurately capture diverse land cover types. Additionally,

the attention mechanism shows promising results by focusing on the most relevant features for building segmentation maps.

TABLE I: Brief literature summary concerning models used in semantic segmentation of satellite imagery

Article	Proposed Method	IoU Score
Ming Wu et al. [8]	Attention Dilated LinkNet 34	64.73
	Attention Dilated LinkNet 50	64.79
	Attention Dilated LinkNet 101	63.37
K. Chaurasia et al. [7]	U-Net	51.00
	PSP-Net	42.00
H. Ghandorh et al. [14]	U-Net	80.32
	SegNet	77.74
S. Seferbekov et al. [10]	FPN	49.30
T.P. Singh et al. [1]	FCN-8s	50.28
	U-Net	78.62
	SegNet	70.60
	DeepLab	76.65
	PSP-Net	71.69

III. DATASET DESCRIPTION

The Mohammed Bin Rashid Space Centre (MBRSC) satellites provided the aerial photos of Dubai that made up the dataset used in this work. It fully captures the urban environment of Dubai with 72 photos arranged into six larger tiles. As seen in Fig. 2, each image tile has matching mask tiles. Building, Land (unpaved area), Road, Vegetation, Water, and an additional Unlabeled class are among the semantic segmentation labels assigned to each image. To increase convenience, the dataset offers clear color representations for every class.

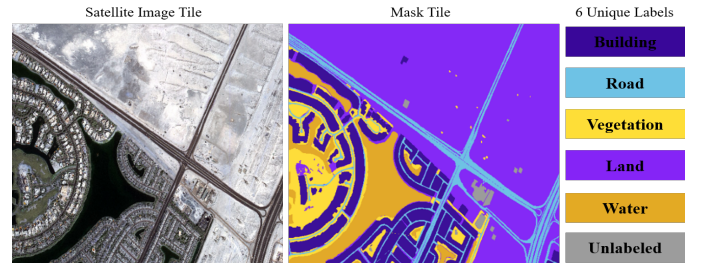


Fig. 2: Satellite image tile and its corresponding mask tile containing six unique labels.

IV. PROPOSED METHODOLOGY

The proposed architecture for residual U-Net-based semantic segmentation of satellite images is illustrated in Fig. 5.

A. Image Pre-processing

The image and mask tiles were too large to be directly used in the model, so they needed to be preprocessed, as shown in Fig. 3, to reduce memory consumption and ensure compatibility with the model's input size.

- **Patchification:** The images are split into smaller patches using the Patchify library, based on a specified patch cell size. This step is used to reduce the memory consumption and computational load, making the large images manageable for the model. It also helps in focusing the model's attention on smaller, more relevant regions of the

image, improving processing efficiency and segmentation accuracy.

- **Normalization:** Pixel values are normalized using Min-MaxScaler to a range of [0, 1] which ensures model stability during semantic segmentation, enabling consistent influence from all pixels and compatibility with the range-dependent loss and activation functions.
- **Patch Size Alignment:** All image tiles and mask images are adjusted to have dimensions that are multiples of the chosen patch size. Each patch is converted into a Numpy array, normalized, and extraneous dimensions are removed.
- **Cropping for Uniformity:** Images with varied sizes are cropped to the nearest size divisible by 256, accommodating both large and small images.

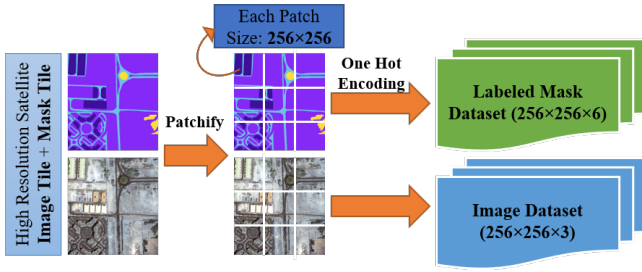


Fig. 3: Preprocessing of satellite images and their corresponding masks.

- **Patch Dimensions:** Images are further subdivided into patches of 256x256x3.
- **Label Encoding:** Hex colors of spatial features are converted to RGB, then mapped to integer labels from 0 to 5, representing different spatial features within the masks.
- **Data Splitting:** The dataset is divided into 75% for training and 25% for testing, with a fixed random state of 42 to ensure reproducibility across experiments.

B. Encoder Path with Residual Blocks

The model receives input satellite images with spatial dimensions of 256x256x3. Each step in this model's encoder path increases the depth of feature maps while decreasing spatial dimensions, capturing increasingly intricate patterns. Input is initially downsampled using convolutional layers, batch normalization, and ReLU activation functions, then fed to residual blocks to preserve information. After that, it uses maxpooling to reduce the spatial dimension i.e. in the first encoder block the dimension was 256x256x16, after maxpooling it becomes 128x128x16. The residual block mechanism allows for smoother training by creating shortcut connections that bypass the main convolutional operations. The block takes an input, processes it through two consecutive convolutional layers, each followed by a batch normalization and ReLU activation function, and then adds the input back to the output. This addition helps the network learn "residual" functions or

the changes that should be applied to the input, rather than learning a completely new transformation.

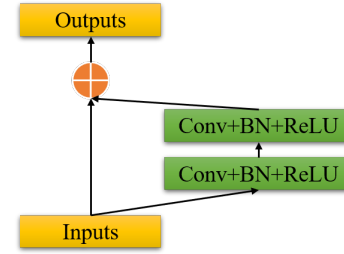


Fig. 4: Mechanism of Modified Residual Blocks.

In Fig. 4 the first residual block, the input image (e.g., 256x256x3) goes through two convolution layers, which transform the pixel values into feature maps (e.g., from 256x256x3 to 256x256x16). A shortcut connection skips these convolutions, passing the original input directly to the output. The two results one from the convolutions and one from the shortcut are added together. This means the network doesn't need to completely change the input but can focus on learning the difference (or residual) between the input and desired output. The final result has the same size as the input (256x256x16) but contains richer information.

C. Decoder Path

After downsampling, the decoder path recovers the spatial resolution of features by upsampling using up-convolution. There is a connection between the encoder and decoder called skip-connection, which preserves essential high-resolution features from the encoder so that information does not get lost during the upsampling process. Starting with the smallest feature map from the bottleneck, each decoder layer applies an upsampling operation to double the spatial resolution (e.g., from 32x32x256 to 64x64x128). At each step, the upsampled feature map is concatenated with the corresponding feature map from the encoder (e.g., the 64x64x128 map from the encoder path is joined with the upsampled map). This process is repeated until it reaches the final output layer where 1x1 convolution is done along with the softmax activation function to get multi-class segmentation output (256x256x6).

D. Attention gate

The attention gate mechanism used in the model helps focus on the most relevant features from the encoder path when concatenating them with the decoder features, improving segmentation accuracy. Given an input feature map(x) from the encoder (e.g., 256x256x16) and a gating signal(g) from the decoder (e.g., 128x128x32), the attention gate aligns these features to highlight important regions.

- **Linear Transformations:** Both x and g are first linearly transformed by 1x1 convolutions:

$$\theta_x = W_x * x, \quad \phi_g = W_g * g \quad (1)$$

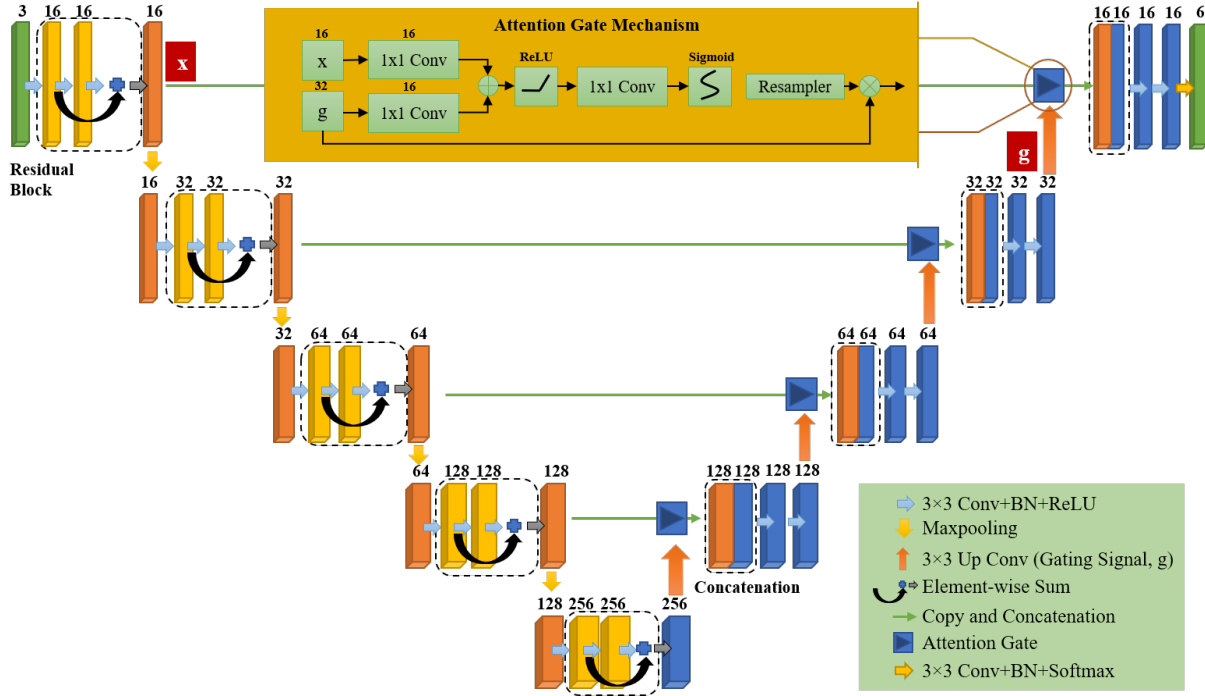


Fig. 5: Proposed model Residual U-Net with Attention Gate Mechanism for semantic segmentation in satellite images.

where θ_x has the same spatial resolution as g . (e.g., $128 \times 128 \times 32$).

- **Element-Wise Sum and Non-Linearity:** The transformed encoder features θ_x and decoder gating signal ϕ_g are then combined using an element-wise sum and passed through a ReLU activation:

$$f = \text{ReLU}(\theta_x + \phi_g + b) \quad (2)$$

This helps focus the features from x that align with g .

- **Attention Coefficients:** The output f is passed through a 1×1 convolution followed by a sigmoid activation to produce attention coefficients α , a map with values between 0 and 1 that scales the encoder features:

$$\alpha = \sigma(W_f \cdot f) \quad (3)$$

where α has the shape $128 \times 128 \times 1$.

- **Resampling and Application:** Finally, α is upsampled to match the dimensions of x (if needed), and element-wise multiplied with x to retain only the important parts:

$$\hat{x} = \alpha \cdot x \quad (4)$$

After that, normal U-net flow continues.

Thus, the attention gate mechanism selectively emphasizes the encoder features that are spatially aligned with relevant decoder features, enhancing feature propagation and IoU score in the final output.

V. RESULT ANALYSIS

A. Evaluation Metrics

We evaluate the quality of output segmentation maps using Intersection over Union score (IoU), accuracy, precision, recall, and f1 score.

1) *Intersection over Union (IoU):* The Intersection over Union (IoU) is one of the most important metrics for evaluating segmentation accuracy. It calculates the difference between predicted and ground truth segmentation maps by comparing their intersection to their union. The mathematical expression for IoU is as follows:

$$IoU = \frac{\text{Area of Overlap (Intersection)}}{\text{Area of Union}} \quad (5)$$

B. Performance Comparison of Models

To thoroughly assess semantic segmentation models used in satellite image processing, we compared our proposed method with other implemented methods: Modified U-Net, Attention U-Net, U-Net with- ResNet34, DenseNet121, and VGGNet16 backbones.

TABLE II: Performance comparison of different U-Net models on the test set

Models	Accuracy	IoU	Precision	Recall	f ₁ -score
Modified U-Net	83.66%	66.09%	84.81%	82.51%	83.65%
U-Net + ResNet34 Backbone	86.51%	73.51%	87.29%	85.85%	86.56%
U-Net + DenseNet121 Backbone	88.20%	74.30%	89.20%	86.87%	88.05%
U-Net + VGGNet16 Backbone	91.26%	78.46%	93.27%	89.33%	91.24%
Attention U-Net	89.66%	79.59%	90.17%	89.21%	89.69%
Residual Attention U-Net	92.33%	81.07%	93.89%	90.77%	90.01%

Table II comprehensively analyzes the performance indicators derived from the experiments. The Residual Attention U-Net model demonstrated exceptional accuracy from the table compared to other U-Nets, as evidenced by its accuracy and IoU score of 92.33% and 81.07% respectively. This model utilizes an attention gate in the decoder path to provide an accurate segmentation output map and residual blocks in the encoder path to extract features more proficiently across different classes.

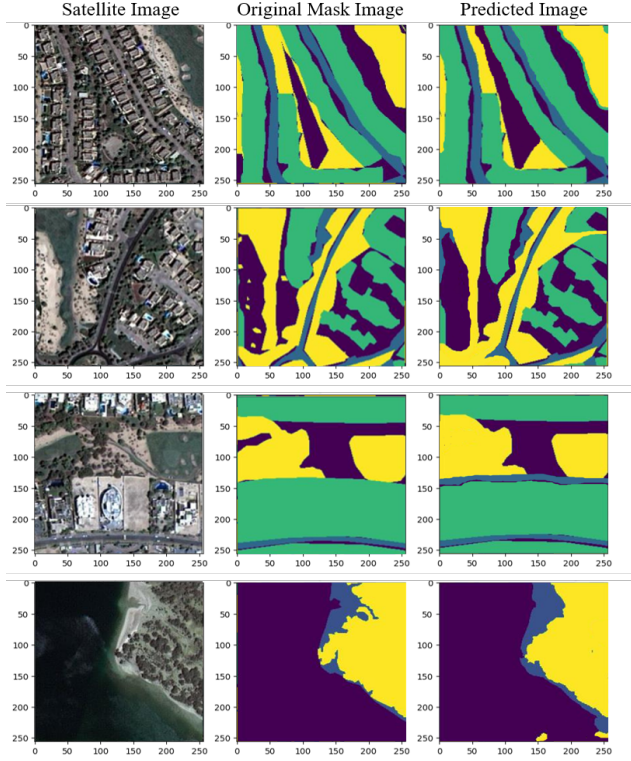


Fig. 6: The predicted semantic segmentation output using Residual Attention U-Net.

From Fig.6, it is displayed the segmentation results generated by our proposed model, Residual Attention U-Net. From the visual analysis, it can be seen that our model has constructed an almost perfect segmentation output while properly detecting all the classes.

C. Class-wise Performance Comparison

The class-wise performance analysis for the Residual Attention U-Net shown in Table III reveals that the model performs well in segmenting different classes within satellite images.

TABLE III: Performance comparison of different classes using Residual Attention U-Net

Classes	Accuracy	IoU
Water	87.30%	75.89%
Land	94.18%	79.98%
Road	85.00%	67.54%
Building	90.87%	73.42%
Vegetation	89.56%	76.78%

The Land class achieves the highest IoU score of 79.98% with a high accuracy of 94.18%, indicating excellent boundary delineation and feature extraction. The Road class, however, has a relatively lower IoU score of 67.54%, suggesting that it may be more challenging for the model to segment. The comparison results are visually illustrated in Fig. 7.

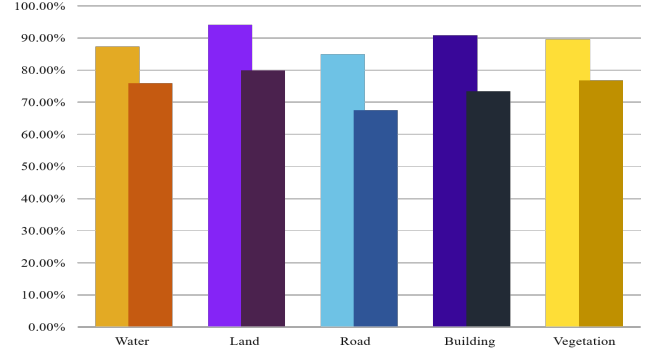


Fig. 7: Comparative analysis of accuracy and IoU score of five classes for Residual Attention U-Net.

D. Comparison of Related Works

We have developed a residual U-Net model with an attention mechanism, as our proposed model. Its performance is compared to the methods used by several researchers in table IV. The U-Net architecture was modified in the

TABLE IV: Comparative analysis of the proposed method with the existing models based on Accuracy and IoU score.

Methods	Accuracy	IoU
U-Net+InceptionResNetV2 [17]	87.00%	-
DensePlusU-Net [18]	86.11%	-
FCN [15]	71.40%	61.80%
CNN [15]	79.40%	70.20%
DeepLabV3 [15]	81.50%	71.90%
Dense+U-Net [15]	81.76%	72.43%
SA-SC U-Net [15]	87.34%	77.45%
RU-Net [16]	86.68%	48.42%
SEU-Net [16]	88.49%	49.00%
SERU-Net [16]	87.06%	47.81%
Residual Attention U-Net (Proposed)	92.33%	81.07%

works of authors cited in [17] and [18] by changing its encoder path using pre-trained models. Similarly, Aburaed et al. [16] incorporated SE blocks, often referred to as attention mechanisms, into the bottleneck section of the U-Net. These techniques prioritize improving feature extraction rather than achieving precise segmentation outputs, which is essential for semantic segmentation. The technique described in reference [15] combines self-attention and separable convolution which yielded a praiseworthy IoU score, but it also resulted in notable attenuation in the resulting segmentation map. Our proposed method, in contrast, achieves a higher level of overall accuracy of 92.33% and an IoU score of 81.07%, which are the greatest among the existing methods described in this context which is represented through a visual comparison in Fig.8. Enhancements to skip connections through attention gates and improvements to encoders with residual blocks contributed to

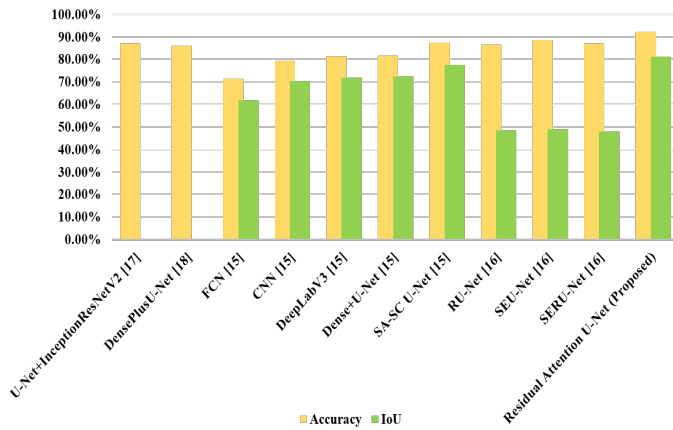


Fig. 8: Comparative analysis of the proposed method with other research approaches.

this success. Skip connections aid in reconstructing segmented outputs, while residual blocks facilitate training by bypassing unnecessary transformations, allowing complex features to be learned.

VI. CONCLUSION AND FUTURE WORK

By combining residual blocks and attention processes, the Residual Attention U-Net model in this work enhances the status of satellite image segmentation. By prioritizing important areas of the image, this combination not only improves feature extraction but also increases segmentation accuracy and resilience across various land cover classes. The Residual Attention U-Net continuously performed better with higher IoU and accuracy than both regular and modified U-Net models, successfully capturing features in intricate landscapes. This study emphasizes how segmentation models can be made more adaptive to high-resolution satellite data by integrating attention mechanisms and residual learning. To track environmental changes or modify the architecture for real-time applications in resource-constrained contexts, future research could investigate expanding this method to multi-temporal datasets. The model's utility in satellite-based monitoring and analysis activities could also be improved by adding more classes or integrating it with other remote sensing technologies.

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